

Dr. Ram Kumar



CONTACT INFORMATION	Department of Mathematics, Amity University, Gwalior, India	+91-9709775959 ramkumarbhu1991@gmail.com
PROFESSIONAL EXPERIENCE	Assistant Professor-II, Amity university, Gwalior, MP, India September 2021-Present. Assistant Professor, SR Group of Institutions, Jhansi, UP, India July 2019-September 2021.	
Research Interests	Number Theory, Algebraic Geometry, Elliptic Curves.	
Education	Indian Institute of Technology (ISM), Dhanbad-826004, Jharkhand, India Ph.D., Mathematics, June 2020. • Thesis Topic: <i>Hypergeometric Functions over Finite Fields and Their Connection to Algebraic Curves</i>, • Advisors: Dr. P. K. Kewat Banaras Hindu University, Varanasi-221005, Uttar Pradesh, India M.Sc., Mathematics, July 2013. Lucknow University, Lucknow-226001, Uttar Pradesh, India. B.Sc., Mathematics, Physics and Geology, July 2011.	
List of Publications	1. Hyperelliptic Curves and its Applications, AIP Conference Proceedings, 1897(1) (2017) pp 020003. Indexed: Scopus 2. Hypergeometric Functions and Algebraic Curves $y^e = x^d + ax + b$, Journal of Ramanujan Math. Soc. 34, No.3(2019) 325-342 (with P. K. Kewat). Indexed: Web of Science(SCIE), Scopus 3. Generalizations of Jacobsthal Sums and hypergeometric series over finite fields, The Ramanujan Journal (2021). 10.1007/s11139-020-00364-w(with P. K. Kewat). Indexed: Web of Science(SCIE), Scopus 4. Some Supercongruences Related to Truncated Hypergeometric Series, The Ramanujan Journal (2023) 157-173. https://doi.org/10.1007/s11139-022-00625-w(with P. K. Kewat). Indexed: Web of Science(SCIE), Scopus	

	5. Type I and Type II codes over the ring $\mathbb{F}_2 + u\mathbb{F}_2 + u^2\mathbb{F}_2 + u^3\mathbb{F}_2$, Ars Combinatoria (2025) 113-121. 10.61091/ars163-08 (with Ankur). Indexed: Web of Science(SCIE), Scopus 6. A Novel Approach to Attaining Initial Basic Feasible Solution of Transportation Problem (accepted)(2026) Indexed: Scopus.
Submitted to Journal	1. Some Special Values of Gaussian Hypergeometric Series over Finite Fields (under review).
Research project submitted	1. Submitted a research proposal to NBHM titled “Point Counting on Algebraic Curves through Gaussian Hypergeometric Functions” (December 2025).
Awards/ Scholarship	● Junior Research Fellowship July 2014–July 2016. ● Senior Research Fellowship July 2016–July 2019. ● Qualified CSIR-NET in Mathematics Dec 2014. ● Qualified GATE in Mathematics Mar 2015. ● Foreign Travel Support from ITS SERB Jan 2018.
Academic Events Organized	● Received a grant of Rs. 3,40,000 from MTTS Trust (funded by NBHM) for organizing an InitMath Camp, conducted from November 3–8, 2025. ● Received a grant of RS 70,000 from MPCST to organize a Short-Term Training Program scheduled from July 28 to August 1, 2025. ● Organized One day Workshop on “Latex” in the Department of Applied Sciences & Humanities, S R Group of Institutions, Jhansi, India. Feb 10, 2020. ● Organized A National Webinar on “Mathematical Analysis of Air-Flow Cooling in Modern Data Centres” in the Department of Applied Sciences & Humanities, S R Group of Institutions, Jhansi, India. June 29, 2020. ● Organized International Workshop on “Recent Trends in Applied Mathematics for Engineering & Technology” in the Department of Applied Mathematics, Amity University Madhya Pradesh, India. August 05-07, 2024. ● Organized National Workshop on “LaTeX Training: Writing Research Paper Beautifully” in the Department of Applied Mathematics, Amity University Madhya Pradesh, India. March 06-07, 2025.
Conference/ Presentations	● Presented Paper on Some Properties of Hypergeometric functions over Finite Fields in the National Conference Recent Advances in Science and Engineering (RASE 2016) held in Indian Institute of Technology (ISM), Dhanbad, India. Mar 28–29, 2016.

	Presented Paper on Hyperelliptic Curves and its Applications in the International Conference Recent Advances in Mathematical Sciences and its Applications (RAMSA 2017) organized by Jaypee Institute of Technology, New Delhi, India. Dec 12–14, 2017.
Conference/ Workshop Participant	- Participated in the 80th Annual Conference of Indian Mathematical Society held in Indian Institute of Technology (ISM), Dhanbad, India. Dec 27–30, 2014. - Attended Advanced Instructional School(AIS) in Analytic Number Theory organized by ATM School National Centre for Mathematics, A joint Centre of IIT Bombay and TIFR, Mumbai held in Kalinga Institute of Industrial Technology (KIIT), Bhubneshwar, India. June 01–20, 2015. - Participated in the 7th National Conference on Wave Mechanics and Vibrations (WMVC) organized by the Indian Institute of Technology (ISM), Dhanbad, India in association with Von Karman Society for Advanced Study and Research in Mathematical Sciences, Jalpaiguri, West Bengal, India. Dec 21–23, 2015. - Participated in the workshop on Communications and Multimedia Security organized by Department of Computer Science and Engineering Indian Institute of Technology (ISM), Dhanbad, India Feb 9–10, 2017. - Attended CIMPA Research School on Explicit Number Theory organized by CIMPA Research School held in The University of Witwatersrand, Johannesburg, South Africa. Jan 08–19, 2018. - Participated in the International Conference on Number Theory organized by Indian Institute of Science Education and Research, Thiruvananthapuram, India. Mar 11–13, 2019. - Participated in the Short Term Course on Matrix Computations and its Applications in system, signal, and Control problems conducted as a part of quality improvement program, organized by the Dept. of Mathematics, IIT, Indore, India. Feb 16–21, 2021. - Participated in the Faculty Development Programme on Research Ethics, organized by the Amity University, Gwalior, India. Jan 07–13, 2022. - Participated in the Teacher’s Enrichment Workshop on “Algebra, Number Theory and Graph Theory”, organized by the Amity University Haryana in association with National Centre for Mathematics (Advanced Training in Mathematics), India. July 04–16, 2022.
Administration Responsibility	IQAC Committee Member IIC Committee Member
Hardware and Software Skills	Computer Programming: C, C++, Basic Knowledge of Mathematica,

Sage, and other
Operating System: Linux, Window, Mac
Other: Latex, Ms Office, and other

Teaching
Experience

- Matrices
- Complex Analysis
- Statistics
- Differential Equations (Ordinary & Partial)
- Numerical Analysis

References

Dr. Pramod Kumar Kewat
Associate Professor Phone:+91-9471191836
Department of Mathematics & Computing E-mail:pramodk@iitism.ac.in
Indian Institute of Technology (ISM),
Dhanbad-826004, Jharkhand, India

Dr. Rupam Barman
Professor Phone:+91 (0)361 258 2637
Department of Mathematics E-mail:rupam@iitg.ac.in
Indian Institute of Technology Guwahati,
Guwahati-781039, Assam, India

Dr. Daya Ram Sahu
Professor Phone:+91-9795371821
Department of Mathematics E-mail:drsahudr@gmail.com
Banaras Hindu University,
Varanasi-221005, Uttar Pradesh, India.



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